

Characterization, Calibration, and Estimation Pipelines for the First Compact 3D Sonar

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Abstract—Over the past decade, the adoption of compact 3D range sensors, such as LiDAR, has catalyzed the development of robust state-estimation pipelines, establishing them as standard modalities for aerial, ground, and space autonomy. However, the rapid attenuation of electromagnetic waves in underwater environments has traditionally limited visibility-independent sensing to acoustic sensors, which typically yield 2D data with significant vertical ambiguity. This paper presents, to the best of our knowledge, the first study examining the performance and capabilities enabled by the recent introduction of the first compact 3D sonar. To this end, we provide an experimental characterization of acoustic responses across various materials and propose a structure-based calibration procedure for estimating the extrinsic transformation between the 3D sonar and a camera. Furthermore, we provide Simultaneous Localization and Mapping (SLAM) pipelines validated through deployments in underwater cave systems and other geometrically and acoustically challenging environments. Our results demonstrate the capacity of 3D sonars to capture consistent spatial information for detailed reconstruction and localization over trajectories spanning hundreds of meters. Finally, we highlight persistent challenges related to acoustic propagation and release our datasets publicly to support community research: <https://github.com/AutonomousFieldRoboticsLab/Compact-3D-Sonar-Datasets-.git>.

I. INTRODUCTION

Mapping underwater environments is time-consuming, labor-intensive, and inherently hazardous. Overhead and cluttered spaces, such as shipwrecks and underwater caves, present significant entanglement risks; cave diving alone has resulted in over 350 fatalities since 1969 in the United States [1], [2]. Despite these dangers, mapping these environments remains critical; for example, Karst aquifers (underwater caves) provide 25% of the world’s potable water [3].

A primary challenge in underwater mapping is limited visibility. Even in relatively clear water, visibility typically degrades after a few meters, making feature extraction extremely challenging. Furthermore, limited illumination and frequency-dependent light attenuation [4], [5] cause most

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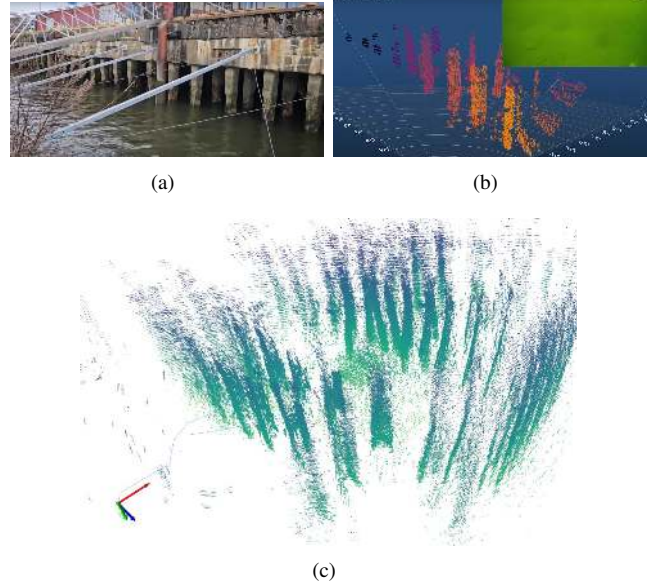


Fig. 1: **3D sonar map reconstruction** in a pier piling environment with limited camera visibility. (a) Pier piling environment, (b) single 3D sonar scan and camera visibility, and (c) reconstructed point cloud results.

vision-based odometry and SLAM systems to underperform [6], which in turn compromises autonomous navigation pipelines that rely upon them [7].

Conversely, while acoustic sensors are popular for their visibility independence, they suffer from inherent limitations. A major drawback is their high angular uncertainty; for instance, the Ping360 from Blue Robotics [8], a 2D mechanical scanning sonar, has a vertical beam opening of 25° , which introduces substantial directional ambiguity. Reducing this uncertainty requires narrower beams, but this comes at the expense of range. For example, the Imagenex pipe profiling sonar [9], another mechanical scanning sonar, achieves a 1.4° beam but is limited to an effective range of 6 meters. 3D sonars may resolve these constraints – as demonstrated in preliminary work utilizing probabilistic scan matching [10] – yet the SWaP-C (size, weight, power, and cost) requirements of earlier devices have historically hindered their adoption on small-scale underwater platforms. Consequently, data collection has largely been limited to “fly-over” maneuvers in open-water environments.

In contrast, newly introduced 3D sonars, such as the Water Linked Sonar 3D-15 [11], are compact, providing a LiDAR-like point cloud (as shown in Fig. 1), without prohibitive logistical overhead. Its form factor enables deployment in complex, cluttered, and overhead environments, facilitating

seamless integration into popular underwater platforms.

This paper evaluates the capabilities of the Sonar 3D-15 through field trials across diverse environments, including a low-visibility dock, a submerged container, a swimming pool for material characterization, a cavern, and a 700 m trajectory within a cave system. With the exception of the dock and the container, visual, inertial, and water depth measurements were collected at the same time. This enabled the development of new calibration procedures, the introduction of SLAM pipelines that integrate such sensors, the collection of dedicated datasets, and the characterization of this novel acoustic-based 3D range sensor.

In summary, the main contributions of this paper are:

- 1) Characterization of the Sonar 3D-15 performance across different material types and surface properties.
- 2) A novel calibration strategy for estimating the extrinsic parameters between the camera and sonar.
- 3) A loosely coupled integration of sonar and VIO into a common estimation process, showing their complementary sensing capabilities.
- 4) The release of several datasets, including sonar, visual, inertial, and water depth data, made publicly available.

II. RELATED WORK

1) *Underwater SLAM*: SLAM is foundational for underwater robot autonomy, providing the necessary state estimation and environmental perception for reliable control and planning. To navigate the unique challenges of aquatic environments, a wide range of sensor configurations have been explored, each tailored to the specific challenges of different environments and tasks. Earlier approaches utilized imaging sonar and Doppler Velocity Logs (DVL) for harbor surveillance [12]. More complex systems, like the Sunfish vehicle [13], [14], integrated multibeam sonar, fiber-optic gyroscope (FOG), Inertial Measurement Unit (IMU), DVLs and pressure sensors for cave exploration and mapping. Visual sensing is often restricted by short operating ranges due to light attenuation, scattering, and dynamic visual disturbances like “marine snow”. To mitigate these issues, Visual-Inertial SLAM (VI-SLAM) is frequently augmented with acoustic or pressure data. For instance, SVIn2 [15] combined visual-inertial sensing with a pipe profiling sonar, and pressure sensor data, extending the framework of OKVIS [16], to map underwater caverns and shipwrecks; later extending to incorporate magnetometer data in [17]. More recently, Turtlmap [18] relied on DVL, inertial, and pressure sensors for localization in low-texture environments and a depth camera for mapping, while [19] utilized side-scan sonar (SSS) with dead-reckoning to address localization in open-water settings.

While proprioceptive sensors, such as IMUs, DVLs, and magnetometers, provide valuable localization cues, they lack the dense environmental awareness required for 3D mapping. Traditional acoustic sensors such as side-scan, and imaging sonars are resilient to turbidity but introduce their own complexities [20]. These low-dimensional sonars suffer from geometric distortion and capture only 2D representations,

leading to 3D positional ambiguity and elevation angle degradation. Recent approaches [21] fuse orthogonal imaging sonar data to resolve this ambiguity, enabling large-scale underwater localization and 3D reconstruction. Parallel to these efforts, earlier works on 3D Sonars providing point cloud data [10], [22], [23] explored underwater mapping and surveying using industrial-grade Echoscope 3D sonar with probabilistic scan registration. These studies demonstrated the promise of 3D sonar for underwater mapping, but sonar-only systems remain vulnerable to sparse and noisy point cloud returns, as well as multipath effects, which can degrade scan association and reduce estimation robustness in challenging environments. In this context, the Water Linked Sonar 3D-15 is a compact sonar provides a point cloud with a sensing range of up to 15 m. This direct 3D sensing largely constrains vertical uncertainty, though the sensor can still suffer from uneven ensonification on specular or smooth surfaces. Here, we evaluate a sonar-only approach in low visibility conditions and explore fusion with Visual-Inertial sensors in environments with satisfactory visibility.

2) *3D Sonar-Camera Extrinsic Calibration*: Calibrating the Sonar 3D-15 extrinsic parameters with a visual sensor appears conceptually similar to LiDAR-camera calibration. Existing LiDAR-camera calibration approaches are broadly categorized into *target-based* and *targetless* methods. In camera-point cloud target-based calibration, prior work [24] has explored a variety of targets, including 2D ArUco markers, checkerboards, and 3D objects (e.g., boxes, spheres, or other structured geometries). Targetless methods instead exploit scene geometry, such as structural regularities [25] or line-plane correspondences [26]. However, conventional flat calibration targets are not reliably perceived by 3D sonar due to specular reflections and limited resolution. Meanwhile, targetless methods that rely on stable geometric structures (e.g., line-plane constraints) are often difficult to apply in natural underwater environments where such features are scarce. In addition, targetless calibration in confined environments such as swimming pools is particularly challenging because sonar measurements are strongly affected by acoustic reverberation and multipath. Hard pool boundaries reflect acoustic pulses, producing delayed echoes that may be interpreted as returns from incorrect bearings and generate ghost measurements that corrupt the geometric consistency required by targetless methods. Although similar practical challenges have been studied in IMU-LiDAR calibration [27], the method requires specific excitation patterns that are difficult to perform underwater. These limitations motivated the development of our custom 3D sonar-camera calibration approach.

III. SONAR CHARACTERIZATION AND CALIBRATION

To evaluate the performance of the new 3D sonar modality and ensure precise spatial alignment with visual sensors, we first define the experimental hardware configuration. The primary sensor is a preproduction unit of Sonar 3D-15 from Water Linked [11]. The Sonar 3D-15 is an advanced 3D sonar with approximately a 90° horizontal by 40° vertical field of view in navigation mode, the only mode tested and

available at the time of this reporting. The navigation mode provides a range of up to 15 m with an update frequency of 5 Hz. The sensor was rigidly mounted on a BlueROV2 platform [28]. The BlueROV2 is equipped with a BlueRobotics Low-Light HD USB camera, an IMU and a pressure sensor measuring the current depth, thus providing globally consistent measurements along the Z-axis. Furthermore, a GoPro Hero Black (10 and 13) with an internal IMU is mounted next to the sonar, facing forward. Illumination is provided by four 1500-lumen Subsea Light [29] and a 20000-lumen Keldan light [30]. Fig. 2 shows the experimental setup deployed inside a cavern in Florida.

The following details the physical characterization of the sonar's material response and the extrinsic calibration procedure used to relate the sonar and camera frames.

A. Sonar Characterization and Material Response

During the experiments, the sonar operated in low (1.2 MHz) frequency, producing a 0.6° horizontal and 2.4° vertical beamwidth. The returned range measurements have 4 mm resolution. Physically, the Sonar 3D-15 measures 0.12 m width by 0.08 m height by 0.04 m depth and with a submerged weight of 390 g.

To select a suitable material for sonar-camera calibration and assess the performance of the 3D sonar with different surfaces, experiments were conducted to evaluate 3D sonar responses across different materials. Selected materials included those frequently encountered in underwater environments, such as bricks, concrete, PVC pipes, and wood. Additionally, easily available items typically used for calibration were included, such as a glass sphere, popular for 2D sonar calibration, and a metal checkerboard with 3D-printed tag for camera calibration. The camera image and corresponding sonar point-cloud of the test materials are shown in Fig. 3. The experiment was conducted in an indoor swimming pool. To minimize the influence of multipath effects caused by the smooth, reflective pool walls, the test materials were suspended in the center of the water column. To further mitigate the impact of these reflections on the material experiment results, the sensor range was limited to $r = 2$ m.

The results demonstrated that the responses varied with both material type and surface properties at different viewing angles (a snapshot is provided in Fig. 3). While the glass

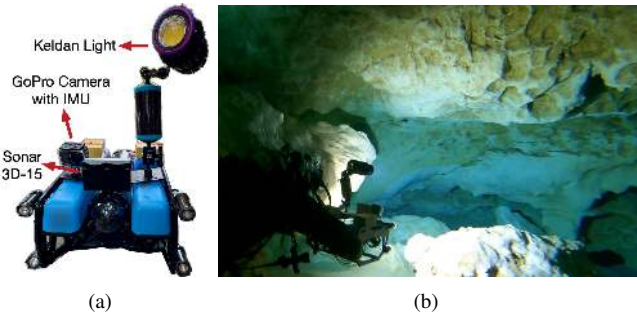


Fig. 2: The customized BlueROV2 setup is shown in (a) and the underwater cave environment during deployment in (b).

sphere and steel calibration board provide strong returns for the 2D sonar, they are not well-suited for 3D sonar calibration due to multipath effects. Regarding surface properties, smooth objects such as PVC pipe produced noisier detections due to reflections, while rougher materials such as concrete yielded more consistent, less noisy measurements. Both spray painted and unpainted concrete produced relatively strong signals, regardless of the surface finish.

B. Calibrating the extrinsic parameters

Based on the material experiments, cinder (concrete) blocks (Fig. 4(a)) were selected due to their rough surface texture, well-defined geometric structure, and convenient availability. Four cinder blocks were placed at the bottom of the water body in a particular arrangement, such that the 16 corners of top surfaces form a distinctive pattern. Vertical surfaces were spray painted to suppress spurious edges and aid corner detection in the camera images. A rectangular mask is used to identify the corners of the four cinder blocks, which are then labeled correspondingly in the visual images. The top surface of the cinder blocks are extracted from sonar point-clouds by identifying the elevated planar regions. As shown in Fig. 4(b), points on these elevated surfaces were subsequently denoised and clustered into four separate 2D surfaces.

After detecting the corner points of the cinder blocks in both the visual images and sonar point cloud, correspondences between them are established to form a 3D-2D association. Define a sonar 3D point as $\mathbf{X} = [x, y, z]^\top$, and a visual point as $\mathbf{p} = [u, v]^\top$. The projection model, with s a scalar scale factor, is given by:

$$s \begin{bmatrix} \mathbf{p} \\ 1 \end{bmatrix} = \mathbf{K} \cdot [\mathbf{R} | \mathbf{t}] \begin{bmatrix} \mathbf{X} \\ 1 \end{bmatrix} \quad (1)$$

The intrinsic matrix $\mathbf{K}$ is obtained through standard camera calibration. The undistorted visual point $\mathbf{p}'$ is defined as $[\mathbf{p}', 1]^\top = \mathbf{K}^{-1}[\mathbf{p}, 1]^\top$. The camera-to-sonar extrinsic $(\mathbf{R}, \mathbf{t})$ calibration problem can then be formulated as a Perspective-n-Point (PnP) problem using 16 brick corner correspondences. We solve this problem using EPnP [31] algorithm. Fig. 5 illustrates the alignment of camera rays passing through 16 block corners in the 2D image plane and their corresponding 3D points, along with the estimated camera-sonar extrinsic.

IV. 3D SONAR ESTIMATION PIPELINES

A. 3D Sonar Scan Registration

Since the point cloud collected by a compact 3D sonar shares many characteristics with solid-state 3D LiDAR, 3D sonar scan registration is performed using the state-of-the-art LiDAR scan registration algorithm, KISS-ICP [32]. KISS-ICP is a point-to-point Iterative Closest Point (ICP) algorithm designed to provide a general solution across different LiDAR platforms and application environments. In this paper, we extend the use of KISS-ICP from various LiDAR configurations to 3D sonar (termed Sonar-odom). Since the sonar point cloud is much sparser than the LiDAR

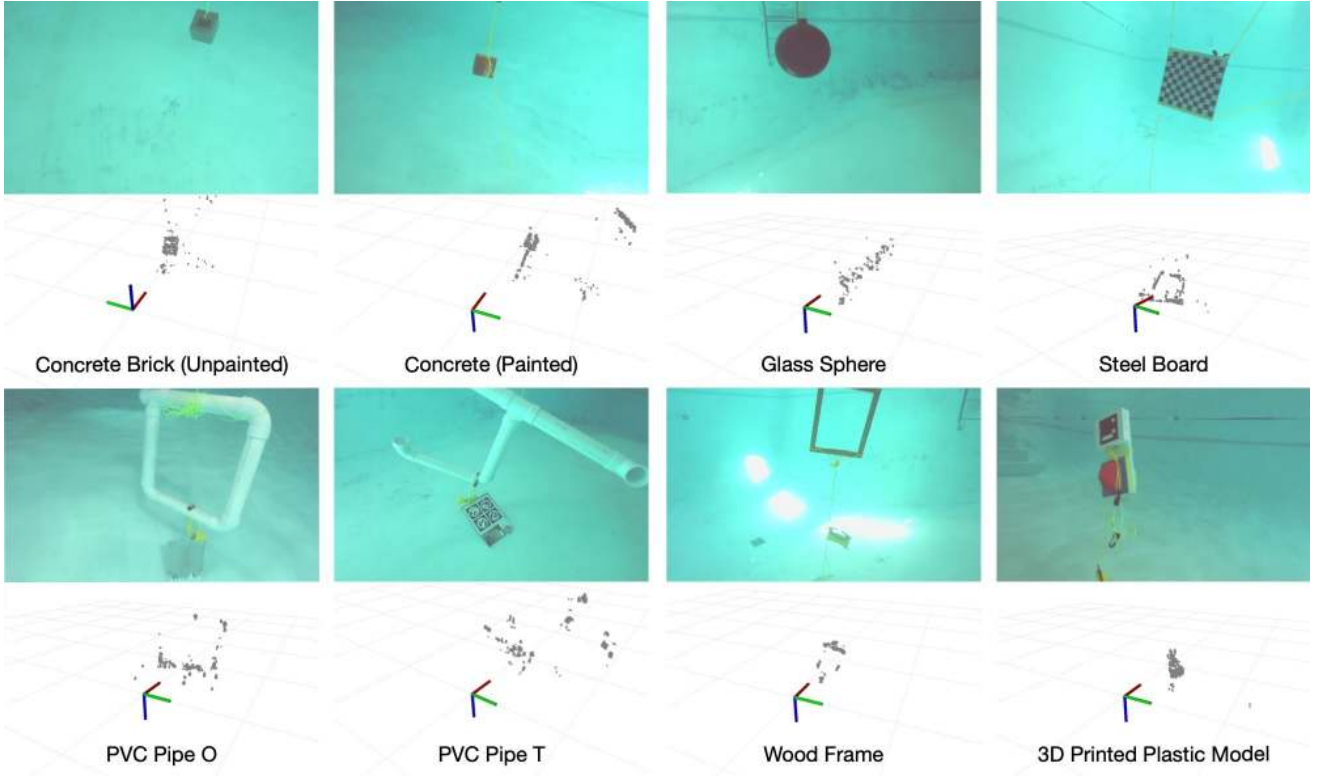


Fig. 3: Different acoustic responses as captured by the Sonar 3D-15 from different materials commonly found in underwater environments or used for calibration.



Fig. 4: Calibration setup: (a) Camera image of the four cinder blocks showing the top-surface borders marked in green and detected corners in blue. (b) Point cloud from the Sonar 3D-15 displaying detected cinder blocks in yellow with borders highlighted in green.

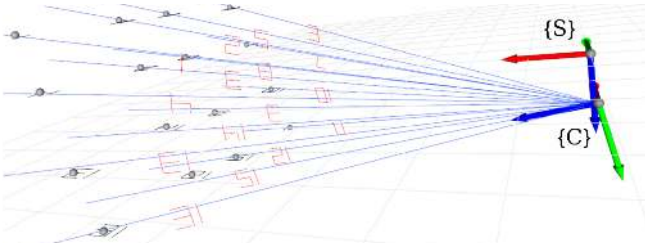


Fig. 5: Rays from the camera projected through 16 cinder block corners in the 2D image plane to their corresponding 3D points. Sonar 3D corner points are annotated in black and visual points in red. The camera frame $\{C\}$ and sonar frame $\{S\}$ are also illustrated in the figure using the calibrated extrinsic transformation.

point cloud, we bypass the down-sampling of the source point cloud and only apply down-sampling on the target point cloud.

KISS-ICP adopts a constant velocity model for generality. At a current timestamp n , the initial guess $\mathbf{T}_{\text{init},n}$ transformation matrix for ICP scan registration is calculated based on

the pose estimates $\mathbf{T}_{n-1}$, $\mathbf{T}_{n-2}$ from previous timestamps:

$$\mathbf{T}_{\text{init},n} = \mathbf{T}_{n-1} \cdot (\mathbf{T}_{n-2}^{-1} \cdot \mathbf{T}_{n-1}) \quad (2)$$

However, ICP is sensitive to the selection of initial guess [33]. Compared to regular 3D LiDARs, which have a wider field of view and higher frequency, the Sonar 3D-15 has a maximum range of only 15 meters with 5 Hz update rate. Furthermore, the complex hydrodynamic conditions in real underwater environments introduce additional uncertainties in the scan registration of the 3D sonar data. To address this, we provide KISS-ICP with a more robust prior derived from the visual-inertial odometry (VIO) of SVIn [15] (termed *Sonar-odom-SVIn-Initial*). The VIO estimation in SVIn is performed at a higher frame rate (30 Hz camera rate). Therefore, we adopt the SVIn odometry from the closest timestamp, $\mathbf{T}_{\text{VIO},n'}$ (with $n' \approx n$), as the initial guess for 3D sonar ICP scan registration. The odometry pose is transformed into the sonar frame, s , using the extrinsic parameters $\Delta\mathbf{T}$ obtained in Section III:

$${}_s\mathbf{T}_{\text{VIO},n'} = \Delta\mathbf{T} \cdot \mathbf{T}_{\text{VIO},n'} \cdot \Delta\mathbf{T}^{-1} \quad (3)$$

The initial guess for the 3D sonar scan registration is then defined as:

$$\mathbf{T}_{\text{init},n} = \mathbf{T}_{n-1} \cdot ({}_s\mathbf{T}_{\text{VIO},n'}^{-1} \cdot {}_s\mathbf{T}_{\text{VIO},n'}) \quad (4)$$

where $n' \approx n-1$ denotes the VIO timestamp closest to $n-1$.

B. 3D Sonar Loop-Closure and Pose Graph Optimization

Due to the relative limited field of view and low update frequency of the sonar, the odometry-only trajectory shows

significant drift, especially over longer durations (Table I). An additional strategy to mitigate drift is loop-closure detection and pose graph optimization, which we call *Sonar PGO*. Similar to LiDAR SLAM [34], we detect loop-closures using a Euclidean distance-based method. Keyframes, denoted as $\mathcal{F} = \{\mathbf{F}_i\}$, are selected to ensure a stable and efficient performance. If the current source keyframe $\mathbf{F}_s$ is geometrically close enough to, and temporally sufficiently separated from a target keyframe, $\mathbf{F}_t$, a potential loop-closure is detected. We then perform scan-to-map matching using the target sub-map, $\mathcal{M}_t = \{\mathbf{F}_{t-w}, \dots, \mathbf{F}_{t+w}\}$, with a window size of $w = 10$.

In pose graph optimization, the sonar pose at each keyframe $\mathbf{F}_i$ is treated as a state $\mathbf{X}_i$. Sonar odometry from scan registration is modeled as a sequential odometry measurement $\mathbf{Z}_{i-1,i}^o$, while the scan-to-map loop-closure is modeled as a loop-closure measurement $\mathbf{Z}_{i,j}^l$. These measurements are added to the graph as factors $f(\mathbf{X}_{i-1}, \mathbf{X}_i; \mathbf{Z}_{i-1,i}^o)$ and $f(\mathbf{X}_i, \mathbf{X}_j; \mathbf{Z}_{i,j}^l)$. The overall factor graph at a given time n , with m detected loop-closures, can be expressed as:

$$f(\mathcal{X}_n; \mathcal{Z}_n) = f(\mathbf{X}_0) \prod_{i=1}^n f(\mathbf{X}_{i-1}, \mathbf{X}_i; \mathbf{Z}_{i-1,i}^o) \prod_{j=1}^m f(\mathbf{X}_i, \mathbf{X}_j; \mathbf{Z}_{i,j}^l) \quad (5)$$

with $\mathcal{X}_n = \{\mathbf{X}_i\}$ denoting the set of all states from time 0 to n , and $\mathcal{Z}_n = \mathcal{Z}_n^o \cup \mathcal{Z}_n^l$ denoting the set containing both odometry measurements $\mathcal{Z}_n^o = \{\mathbf{Z}_{i-1,i}^o\}$ and loop-closure measurements $\mathcal{Z}_n^l = \{\mathbf{Z}_{i,j}^l\}$.

Note we refer to the use of SVIn as an initial guess for sonar registration and loop-closure as the **Fusion approach**. While SVIn originally denotes Sonar Visual-Inertial Navigation, we utilize only its visual-inertial odometry (VIO) component [16], since the sonar sensor used in SVIn2 [15] was a pipe profiling sonar, which differs from our setup.

V. EXPERIMENTAL VALIDATION

We evaluate the proposed methods on four datasets with distinct environmental characteristics:

- *Ginnie Springs Cavern*: A cavern dataset with a larger opening, varying depth, and several narrower passages. The BlueROV2, equipped with a 3D sonar and a GoPro10, was carried by divers along looping trajectories. The dataset was collected over two days: Day I with a powerful Keldan light, and Day II with less powerful lights mounted on the BlueROV2. The Ginnie Springs dataset contains floating particulates in the images, bringing challenges for the state-of-art underwater VIO framework.
- *Peacock Springs Cave*: A long cave trajectory collected using a BlueROV2 equipped with a mounted GoPro13 and carried by cave divers. The divers entered and exited through the same area, resulting in a trajectory with a start-end loop. The Peacock Springs dataset is a long-duration dataset that demonstrates the ability of our system to handle extended localization and mapping missions.
- *Pier Pilling*: A structured, shallow-water dataset featuring repetitive vertical geometries (wood piles), collected using a teleoperated BlueROV2. While geometrically

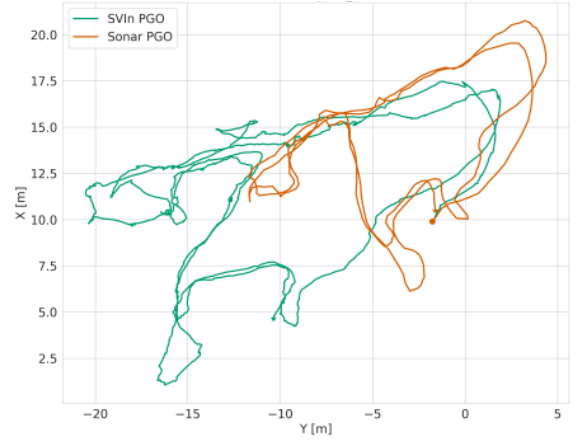


Fig. 6: Comparison of different trajectories produced by the sonar-only approach and SVIn on the Ginnie Ballroom I dataset.

simpler than the cavern environment, it provides a benchmark for evaluating registration stability under wave-induced motion and acoustic multipath.

- *Container*: A dataset of submerged metal tank with an opening. The container dataset represents a challenging environment that includes a steel container. The steel would cause multipath effect for the 3D sonar, making localization and mapping using sonar point cloud even more challenging.

Since the datasets were recorded using multiple devices with unsynchronized clocks, temporal alignments was performed in post-processing by synchronizing the individual data-streams using raw gyroscope measurements.

We compare Sonar SLAM (referred to as Sonar PGO) and our visual-sonar fusion approach (referred to as Fusion PGO) to the state-of-art underwater VIO, SVIn2 (referred to as SVIn PGO).

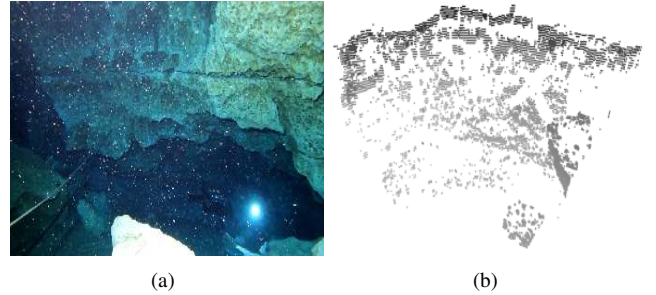


Fig. 7: GoPro camera image (a) and sonar point cloud (b) captured at nearly the same timestamp. The camera image is filled with floating particulates, whereas the sonar point cloud is unaffected by them.

A. Evaluation of start-end re-visitation drift

Fig. 6 presents the comparison between Sonar PGO and SVIn PGO on a start-end loop trajectory, referred to here as Ginnie Ballroom I dataset. All trajectories are aligned using the keyframe poses from first five timestamps of the Sonar PGO with EVO¹, and the starting and ending points are marked by a circle and a star, respectively. Due to the

¹EVO: <https://github.com/MichaelGrupp/evo>

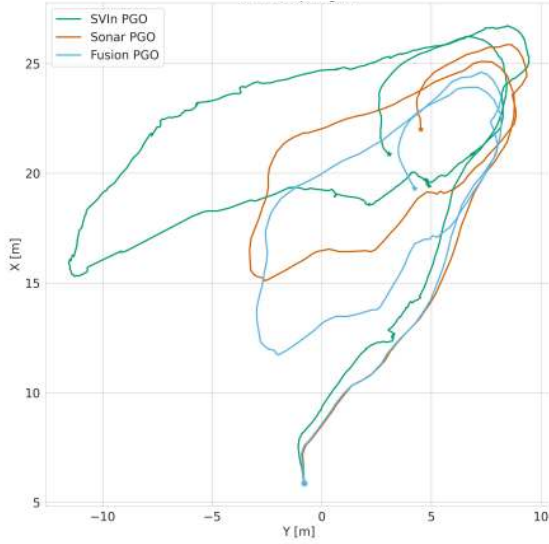


Fig. 8: Trajectories comparison of our Fusion approach, SVIn and Sonar-only on the Ginnie Ballroom II dataset.

presence of floating particulates (Fig. 7), which are detected and misinterpreted as static features, SVIn exhibits a larger start-end re-visitation drift (10.34 m) compared to Sonar PGO (3.31 m). Though raw sonar scan matching suffers from the drift caused by the low sensor update frequency and the constant-velocity motion model used by KISS-ICP, the longer sonar range (15 meters) ensures that enough loop closures are detected, thereby constraining the overall drift and achieving a better start-end re-visitation drift error.

We evaluate the performance of our fusion approach on the datasets collected in Ginnie Ballroom II, and compared it against two configurations, Sonar-only SLAM and SVIn. Fig. 8 shows the trajectories using various configurations. Similarly, all trajectories are aligned using the first five timestamps of Sonar PGO. Incorporating the SVIn prior into Sonar-PGO reduces overall drift compared to both methods on the Ginnie Ballroom II dataset. The vision-based method drifts when there are floating particulates, whereas the sonar remains unaffected by such occlusions and provides reliable longer range measurements. However, sonar-only approaches suffer from registration errors due to low measurement frequency. This drawback is mitigated by the higher-frequency odometry from SVIn, which supplies robust motion priors for scan alignment. By combining both camera and sonar data, the system exhibits lower drift compared to trajectories using only the camera or the sonar.

This finding is apparent also on the Peacock Springs Cave dataset. The re-visitation error for the start-end loop of around a 700-meter trajectory collected in one of the Peacock Springs cave systems is summarized in Table I. The error is calculated as the Euclidean distance between the starting and ending points of the trajectory. The cave entrance has a diameter of approximately 10 meters; therefore, even if the starting and ending points do not coincide, the re-visitation error should be no greater than 10 meters. Our fusion approach achieves a re-visitation error of 6.76 meters, which is the closest to this theoretical bound.

TABLE I: **Re-visitation Error [m]** between the start and end points of the Peacock Springs cave trajectory under different methods and configurations.

SVIn PGO	Sonar Odом	Sonar PGO	Fusion Odом	Fusion PGO
17.02	250.34	21.51	238.71	6.76

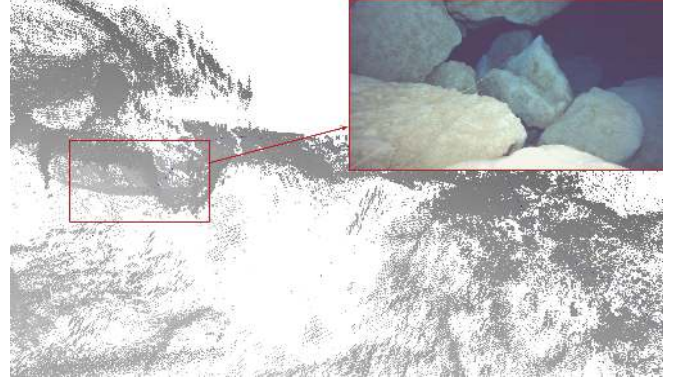


Fig. 9: Detailed point cloud reconstruction of the cave interior from the Ginnie Ballroom I dataset, using a limestone structure as a reference.

B. Dense Point Cloud Reconstruction

1) *Reconstruction of the Cave System:* We also demonstrate the dense reconstruction results of our fusion approach in the challenging environments of the Ginnie Springs cavern (Fig. 9) and the Peacock Springs (Fig. 10 and 11).

The Sonar 3D-15 enables consistent dense reconstruction, as shown in a zoomed-in view of Ginnie Ballroom interior. As mentioned previously, this dataset contains floating particulates that interfere with the performance of SVIn. However, in the sonar reconstruction results, some basic topographic features of the cave environment are perceptible, demonstrating the effectiveness of our fusion and loop closure detection strategy. In Fig. 9, the limestone structure and a nearby narrow passage can be roughly identified in the dense point cloud, corresponding well to the collected image.

The Peacock Spring dataset presents confined, tunnel-like settings, posing significant difficulties for conventional 3D sensing modalities. Although SVIn achieves a relatively low drift rate, as shown previously in Table I, failures in loop closure due to sparse visual features and viewpoint differences, as illustrated in Fig. 11(a), prevent successful loop closure. In contrast, the fusion approach is largely unaffected by these limitations. As shown in Fig. 10, it effectively reconstructs even the narrowest part of the dense structural layout of the confined cave environment. We generate the dense point cloud from sonar point clouds associated with the keyframes in the pose graph optimization. Compared to the existing Cave Atlas maps, which are manually surveyed and contain only the topographic relationships of the caves, the reconstructed point cloud map provides finer structural detail, improved scale accuracy, and a full 3D representation.

2) *Operation in Acoustically Challenging Environments:* The Pier Piling (Fig. 1) and Container (Fig. 12(a)) datasets evaluate the sensor's performance in confined, low-visibility environments. In the Pier Piling dataset, the estimation pipeline performs robustly, yielding a reconstructed point cloud with clearly defined vertical pillars (Fig. 1(c)).

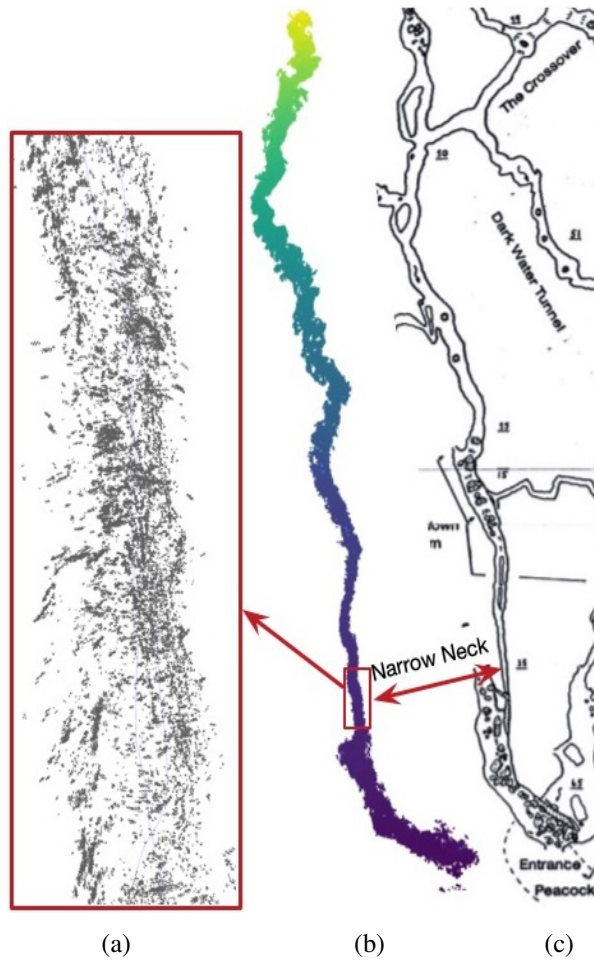


Fig. 10: Results from the Fusion approach on a 700m Peacock Springs cave trajectory: (a) detailed point cloud of the narrow neck, (b) point cloud map of the explored cave system, and (c) the corresponding Cave Atlas map.

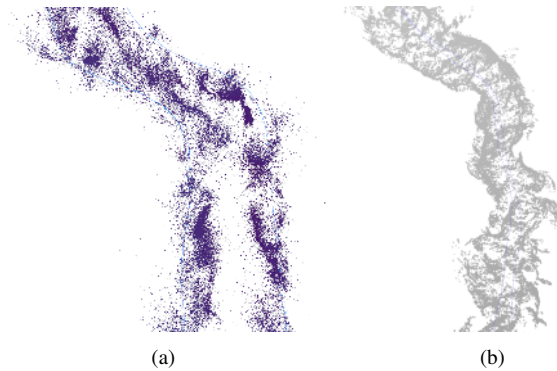


Fig. 11: Comparison of detailed point cloud reconstructions of the same area in Peacock Caves: (a) SVIn PGO and (b) Fusion PGO approach.

As noted in the material characterization, some sonar scans are incomplete due to grazing angles or acoustic absorption, such as at the container door base (Fig. 12(b)). However, geometric features are captured with relative consistency, including the triangular structures at the corners. In the full reconstruction (Fig. 12(c)), the overall shape of the structure is captured well forming very accurately the cylindrical shape of the container, though several artifacts are present.

Specifically, multipath reflections within the confined in-

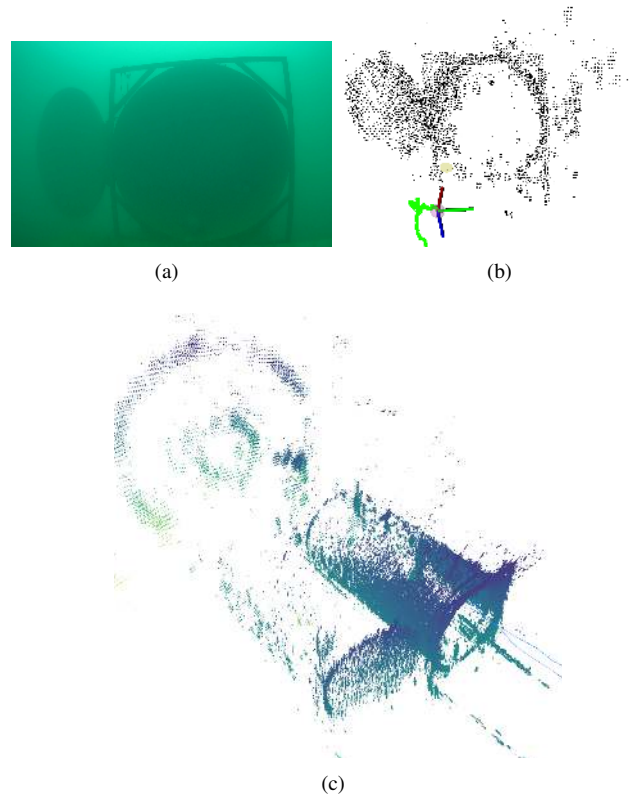


Fig. 12: Reconstruction outliers in acoustically challenging environments. Visibility is shown in (a), sonar scan registration in (b), and the resulting reconstruction in (c). Outliers may occur due to multipath issues.

ternal space generated sparse outliers along the lateral walls at a radial distance consistent with the cylinder's diameter. Moreover, a large interference pattern emerged in the form of two homocentric rings, on the other open-ended side of the cylindrical structure at a similar distance as its length, indicating further multipath inconsistencies in highly reflective, enclosed underwater environments.

VI. LESSONS LEARNED, FUTURE WORK, AND CONCLUSIONS

One of the key observations was the effect of the field of view on the trajectory estimation. While the wider horizontal field of view (90°) maintained feature tracking, the restricted vertical field of view (40°) during operations in the narrow tunnels of the Peacock cave system, FL, resulted in systematic drift over time. More specifically, as only a small patch of the floor and ceiling was visible for large segments of the trajectory, ICP resulted in incorrect matches and incremental drift on the Z-axis; loop-closures on the way back brought the trajectory back to the starting point. The final goal of this work will be to integrate all the available sources of information, acoustic, visual, inertial, water depth, and possibly magnetic field, into a tightly coupled optimization framework. Contrary to SVIn2 [15] where the pipe profiling sonar and the camera did not share a field of view, the Sonar 3D-15 shares the field of view of the camera, simplifying the correlation of features in the observed environment.

The integration of the range measurements into a coherent volumetric map is also part of the future work. Such a development will assist in multi-session mapping where an environment is explored and mapped over several deployments. Commercial work by Sunfish Inc. [14] has generated an example of visualization. Exploring different representations, such as voxels [35], Surflets [36], or learning based Gaussian Splatting like representations in terms of accuracy and ease of integration into the state estimation process will open several directions of future research.

Finally, in challenging environments, as shown in our experiments, typical acoustic issues remain, such as multipath effects. Thus, future research may focus on online corrections through probabilistic map representations.

Overall, this paper characterized a new 3D sonar in different challenging underwater environments for localization and dense reconstruction, providing the foundation for robust underwater mapping and exploration. By presenting a novel calibration strategy and SLAM pipelines and releasing our datasets, we aim to accelerate research into the unique challenges and opportunities of 3D acoustic sensing for enhanced underwater robot autonomy.

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